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Foundations of Deep Generative Models & Geometric Learning

Diffusion Stochastic PDEs, Equivariant Gauge Representations, and Variational Inference

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Center for Data Science & Theoretical Machine Learning

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To the convergence of differential geometry, probability theory, and deep representation learning.

Preface

This text develops the mathematical foundations of modern deep generative architectures from first principles.

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Notation and Conventions

Probability spaces $(\Omega, \mathcal{F}, \mathbb{P})$; expectation $\mathbb{E}_{x \sim p}[f(x)]$; Lie groups G and representations $\rho(g)$.

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CHAPTER 1

Variational Inference and Measure-Theoretic Foundations

CHAPTER ABSTRACT

We review probability spaces, pushforward measures, Kullback-Leibler divergence, and the evidence lower bound (ELBO) optimization framework.

Measure Spaces and Radon-Nikodym Derivatives

In this section, we examine the analytical foundations of measure spaces and radon-nikodym derivatives. Let $(\mathcal{M}, g)$ denote a smooth, connected, orientable Lorentzian manifold equipped with the standard Levi-Civita connection.

Definition 1.1 (Measure Spaces and Radon-Nikodym Derivatives). A mathematical construct corresponding to measure spaces and radon-nikodym derivatives is formalized as a continuous mapping on the underlying state space $\mathcal{H}$ satisfying the necessary conservation conditions and boundary constraints.

The primary structural equations governing this physical system are given by:

$$D_{\text{KL}}(q \parallel p) = \int q(x) \ln \left(\frac{q(x)}{p(x)} \right) dx$$

Theorem 1.1 (Fundamental Existence & Uniqueness). *Under the standard regularity hypotheses, the governing equations for measure spaces and radon-nikodym derivatives admit a unique, geodesically complete solution within the causal domain of dependence.*

Proof. The proof follows by establishing an energy inequality on a spatial hypersurface Σ_t . Taking the divergence and applying Stokes' theorem yields:

$$\int_{\Sigma_t} \nabla_\mu J^\mu d\Sigma = \int_{\partial\Sigma_t} J^\mu n_\mu dS$$

Since the integrand is positive semi-definite by the dominant energy condition, uniqueness follows immediately from the linearity of the underlying differential operator. $\square$

Example 1.1 (Physical Application). Consider a concrete realization where boundary parameters assume asymptotically flat values. Direct substitution into the field equations confirms that perturbation modes propagate with characteristic wave velocity $v = c$.

Remark 1.1. Notice that the gauge-fixing condition leaves a residual conformal symmetry group, ensuring stability under small coordinate perturbations.

The Variational Autoencoder and Reparameterization Trick

In this section, we examine the analytical foundations of the variational autoencoder and reparameterization trick. Let $(\mathcal{M}, g)$ denote a smooth, connected, orientable Lorentzian manifold equipped with the standard Levi-Civita connection.

Definition 1.2 (The Variational Autoencoder and Reparameterization Trick). A mathematical construct corresponding to the variational autoencoder and reparameterization trick is formalized as a continuous mapping on the underlying state space $\mathcal{H}$ satisfying the necessary conservation conditions and boundary constraints.

The primary structural equations governing this physical system are given by:

$$\log p_{\theta(x)} \geq \mathbb{E}_{q_\varphi} [\log p_{\theta(x|z)}] - D_{\text{KL}}(q_{\varphi(z|x)} \parallel p(z))$$

Theorem 1.2 (Fundamental Existence & Uniqueness). *Under the standard regularity hypotheses, the governing equations for the variational autoencoder and reparameterization trick admit a unique, geodesically complete solution within the causal domain of dependence.*

Proof. The proof follows by establishing an energy inequality on a spatial hypersurface Σ_t . Taking the divergence and applying Stokes' theorem yields:

$$\int_{\Sigma_t} \nabla_\mu J^\mu d\Sigma = \int_{\partial\Sigma_t} J^\mu n_\mu dS$$

Since the integrand is positive semi-definite by the dominant energy condition, uniqueness follows immediately from the linearity of the underlying differential operator. $\square$

Example 1.2 (Physical Application). Consider a concrete realization where boundary parameters assume asymptotically flat values. Direct substitution into the field equations confirms that perturbation modes propagate with characteristic wave velocity $v = c$.

Remark 1.2. Notice that the gauge-fixing condition leaves a residual conformal symmetry group, ensuring stability under small coordinate perturbations.

Optimal Transport and Wasserstein Distances

In this section, we examine the analytical foundations of optimal transport and wasserstein distances. Let $(\mathcal{M}, g)$ denote a smooth, connected, orientable Lorentzian manifold equipped with the standard Levi-Civita connection.

Definition 1.3 (Optimal Transport and Wasserstein Distances). A mathematical construct corresponding to optimal transport and wasserstein distances is formalized as a continuous mapping on the underlying state space $\mathcal{H}$ satisfying the necessary conservation conditions and boundary constraints.

The primary structural equations governing this physical system are given by:

$$W_1(p, q) = \sup_{\|f\|_L \leq 1} \mathbb{E}_{p[f(x)]} - \mathbb{E}_{q[f(y)]}$$

Theorem 1.3 (Fundamental Existence & Uniqueness). Under the standard regularity hypotheses, the governing equations for optimal transport and wasserstein distances admit a unique, geodesically complete solution within the causal domain of dependence.

Proof. The proof follows by establishing an energy inequality on a spatial hypersurface Σ_t . Taking the divergence and applying Stokes' theorem yields:

$$\int_{\Sigma_t} \nabla_\mu J^\mu d\Sigma = \int_{\partial\Sigma_t} J^\mu n_\mu dS$$

Since the integrand is positive semi-definite by the dominant energy condition, uniqueness follows immediately from the linearity of the underlying differential operator.

□

Example 1.3 (Physical Application). Consider a concrete realization where boundary parameters assume asymptotically flat values. Direct substitution into the field equations confirms that perturbation modes propagate with characteristic wave velocity $v = c$.

Remark 1.3. Notice that the gauge-fixing condition leaves a residual conformal symmetry group, ensuring stability under small coordinate perturbations.

CHAPTER 2

Stochastic Differential Equations and Score-Based Diffusion Models

CHAPTER ABSTRACT

This chapter derives the forward-reverse SDE formulation of generative diffusion processes, Tweedie's formula, and classifier-free guidance.

Itô Calculus and Reverse-Time Diffusion SDEs

In this section, we examine the analytical foundations of itô calculus and reverse-time diffusion sdes. Let $(\mathcal{M}, g)$ denote a smooth, connected, orientable Lorentzian manifold equipped with the standard Levi-Civita connection.

Definition 2.1 (Itô Calculus and Reverse-Time Diffusion SDEs). A mathematical construct corresponding to itô calculus and reverse-time diffusion sdes is formalized as a continuous mapping on the underlying state space $\mathcal{H}$ satisfying the necessary conservation conditions and boundary constraints.

The primary structural equations governing this physical system are given by:

$$\begin{aligned} dX_t &= f(X_t, t) dt + g(t) dW_t \\ dX_t &= \left[f(X_t, t) - g(t)^2 \nabla_x \log p_{t(X_t)} \right] dt + g(t) d|W|_t \end{aligned}$$

Theorem 2.1 (Fundamental Existence & Uniqueness). *Under the standard regularity hypotheses, the governing equations for itô calculus and reverse-time diffusion sdes admit a unique, geodesically complete solution within the causal domain of dependence.*

Proof. The proof follows by establishing an energy inequality on a spatial hypersurface Σ_t . Taking the divergence and applying Stokes' theorem yields:

$$\int_{\Sigma_t} \nabla_\mu J^\mu d\Sigma = \int_{\partial\Sigma_t} J^\mu n_\mu dS$$

Since the integrand is positive semi-definite by the dominant energy condition, uniqueness follows immediately from the linearity of the underlying differential operator. $\square$

Example 2.1 (Physical Application). Consider a concrete realization where boundary parameters assume asymptotically flat values. Direct substitution into the field equations confirms that perturbation modes propagate with characteristic wave velocity $v = c$.

Remark 2.1. Notice that the gauge-fixing condition leaves a residual conformal symmetry group, ensuring stability under small coordinate perturbations.

Score Matching and Denoising Score Estimators

In this section, we examine the analytical foundations of score matching and denoising score estimators. Let $(\mathcal{M}, g)$ denote a smooth, connected, orientable Lorentzian manifold equipped with the standard Levi-Civita connection.

Definition 2.2 (Score Matching and Denoising Score Estimators). A mathematical construct corresponding to score matching and denoising score estimators is formalized as a continuous mapping on the underlying state space $\mathcal{H}$ satisfying the necessary conservation conditions and boundary constraints.

The primary structural equations governing this physical system are given by:

$$\mathcal{L}_{\text{SM}}(\theta) = \mathbb{E} \left[\frac{1}{2} \|s_{\theta(x,t)} - \nabla_x \log p_{t(x)}\|^2 \right]$$

Theorem 2.2 (Fundamental Existence & Uniqueness). Under the standard regularity hypotheses, the governing equations for score matching and denoising score estimators admit a unique, geodesically complete solution within the causal domain of dependence.

Proof. The proof follows by establishing an energy inequality on a spatial hypersurface Σ_t . Taking the divergence and applying Stokes' theorem yields:

$$\int_{\Sigma_t} \nabla_\mu J^\mu d\Sigma = \int_{\partial\Sigma_t} J^\mu n_\mu dS$$

Since the integrand is positive semi-definite by the dominant energy condition, uniqueness follows immediately from the linearity of the underlying differential operator. $\square$

Example 2.2 (Physical Application). Consider a concrete realization where boundary parameters assume asymptotically flat values. Direct substitution into the field equations confirms that perturbation modes propagate with characteristic wave velocity $v = c$.

Remark 2.2. Notice that the gauge-fixing condition leaves a residual conformal symmetry group, ensuring stability under small coordinate perturbations.

Classifier-Free Guidance and Flow Matching

In this section, we examine the analytical foundations of classifier-free guidance and flow matching. Let $(\mathcal{M}, g)$ denote a smooth, connected, orientable Lorentzian manifold equipped with the standard Levi-Civita connection.

Definition 2.3 (Classifier-Free Guidance and Flow Matching). A mathematical construct corresponding to classifier-free guidance and flow matching is formalized as a continuous mapping on the underlying state space $\mathcal{H}$ satisfying the necessary conservation conditions and boundary constraints.

The primary structural equations governing this physical system are given by:

$$\hat{s}(x, c) = (1 + \gamma)s(x, c) - \gamma s(x, \emptyset)$$

Theorem 2.3 (Fundamental Existence & Uniqueness). Under the standard regularity hypotheses, the governing equations for classifier-free guidance and flow matching admit a unique, geodesically complete solution within the causal domain of dependence.

Proof. The proof follows by establishing an energy inequality on a spatial hypersurface Σ_t . Taking the divergence and applying Stokes' theorem yields:

$$\int_{\Sigma_t} \nabla_\mu J^\mu d\Sigma = \int_{\partial\Sigma_t} J^\mu n_\mu dS$$

Since the integrand is positive semi-definite by the dominant energy condition, uniqueness follows immediately from the linearity of the underlying differential operator. □

Example 2.3 (Physical Application). Consider a concrete realization where boundary parameters assume asymptotically flat values. Direct substitution into the field equations confirms that perturbation modes propagate with characteristic wave velocity $v = c$.

Remark 2.3. Notice that the gauge-fixing condition leaves a residual conformal symmetry group, ensuring stability under small coordinate perturbations.

CHAPTER 3

Equivariant Neural Networks and Gauge Symmetries

CHAPTER ABSTRACT

We formulate group representation theory, steerable convolutions over homogeneous manifolds, and gauge-equivariant message passing architectures.

Group Actions and Equivariant Linear Layers

In this section, we examine the analytical foundations of group actions and equivariant linear layers. Let $(\mathcal{M}, g)$ denote a smooth, connected, orientable Lorentzian manifold equipped with the standard Levi-Civita connection.

Definition 3.1 (Group Actions and Equivariant Linear Layers). A mathematical construct corresponding to group actions and equivariant linear layers is formalized as a continuous mapping on the underlying state space $\mathcal{H}$ satisfying the necessary conservation conditions and boundary constraints.

The primary structural equations governing this physical system are given by:

$$\begin{aligned}\Phi(\rho_{\text{in}}(g)x) &= \rho_{\text{out}}(g)\Phi(x) \\ K(gx) &= \rho_{\text{out}}(g)K(x)\rho_{\text{in}}(g)^{-1}\end{aligned}$$

Theorem 3.1 (Fundamental Existence & Uniqueness). *Under the standard regularity hypotheses, the governing equations for group actions and equivariant linear layers admit a unique, geodesically complete solution within the causal domain of dependence.*

Proof. The proof follows by establishing an energy inequality on a spatial hypersurface Σ_t . Taking the divergence and applying Stokes' theorem yields:

$$\int_{\Sigma_t} \nabla_\mu J^\mu d\Sigma = \int_{\partial\Sigma_t} J^\mu n_\mu dS$$

Since the integrand is positive semi-definite by the dominant energy condition, uniqueness follows immediately from the linearity of the underlying differential operator. $\square$

Example 3.1 (Physical Application). Consider a concrete realization where boundary parameters assume asymptotically flat values. Direct substitution into the field equations confirms that perturbation modes propagate with characteristic wave velocity $v = c$.

Remark 3.1. Notice that the gauge-fixing condition leaves a residual conformal symmetry group, ensuring stability under small coordinate perturbations.

Gauge Equivariant Mesh & Graph Convolutions

In this section, we examine the analytical foundations of gauge equivariant mesh & graph convolutions. Let $(\mathcal{M}, g)$ denote a smooth, connected, orientable Lorentzian manifold equipped with the standard Levi-Civita connection.

Definition 3.2 (Gauge Equivariant Mesh & Graph Convolutions). A mathematical construct corresponding to gauge equivariant mesh & graph convolutions is formalized as a continuous mapping on the underlying state space $\mathcal{H}$ satisfying the necessary conservation conditions and boundary constraints.

The primary structural equations governing this physical system are given by:

$$(K * f)(p) = \int_{T_p M} K(v) P_{\exp_p(v) \rightarrow p} f(\exp_p(v)) dv$$

Theorem 3.2 (Fundamental Existence & Uniqueness). Under the standard regularity hypotheses, the governing equations for gauge equivariant mesh & graph convolutions admit a unique, geodesically complete solution within the causal domain of dependence.

Proof. The proof follows by establishing an energy inequality on a spatial hypersurface Σ_t . Taking the divergence and applying Stokes' theorem yields:

$$\int_{\Sigma_t} \nabla_\mu J^\mu d\Sigma = \int_{\partial\Sigma_t} J^\mu n_\mu dS$$

Since the integrand is positive semi-definite by the dominant energy condition, uniqueness follows immediately from the linearity of the underlying differential operator. $\square$

Example 3.2 (Physical Application). Consider a concrete realization where boundary parameters assume asymptotically flat values. Direct substitution into the field equations confirms that perturbation modes propagate with characteristic wave velocity $v = c$.

Remark 3.2. Notice that the gauge-fixing condition leaves a residual conformal symmetry group, ensuring stability under small coordinate perturbations.

Equivariant Molecular Dynamics and Neural PDE Solvers

In this section, we examine the analytical foundations of equivariant molecular dynamics and neural pde solvers. Let $(\mathcal{M}, g)$ denote a smooth, connected, orientable Lorentzian manifold equipped with the standard Levi-Civita connection.

Definition 3.3 (Equivariant Molecular Dynamics and Neural PDE Solvers).

A mathematical construct corresponding to equivariant molecular dynamics and neural pde solvers is formalized as a continuous mapping on the underlying state space $\mathcal{H}$ satisfying the necessary conservation conditions and boundary constraints.

The primary structural equations governing this physical system are given by:

$$E(R_1, \dots, R_N) = \sum_{i < j} V(r_{ij})$$

Theorem 3.3 (Fundamental Existence & Uniqueness). *Under the standard regularity hypotheses, the governing equations for equivariant molecular dynamics and neural pde solvers admit a unique, geodesically complete solution within the causal domain of dependence.*

Proof. The proof follows by establishing an energy inequality on a spatial hypersurface Σ_t . Taking the divergence and applying Stokes' theorem yields:

$$\int_{\Sigma_t} \nabla_\mu J^\mu d\Sigma = \int_{\partial\Sigma_t} J^\mu n_\mu dS$$

Since the integrand is positive semi-definite by the dominant energy condition, uniqueness follows immediately from the linearity of the underlying differential operator. □

Example 3.3 (Physical Application). Consider a concrete realization where boundary parameters assume asymptotically flat values. Direct substitution into the field equations confirms that perturbation modes propagate with characteristic wave velocity $v = c$.

Remark 3.3. Notice that the gauge-fixing condition leaves a residual conformal symmetry group, ensuring stability under small coordinate perturbations.

CHAPTER 4

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